

FORECASTING ELI LILLY (LLY) STOCK PRICE

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WHY FORECAST ELI LLY?

The Client Scenario

A healthcare-focused investment fund is evaluating a long-term position in Eli Lilly (LLY), driven by growth in its GLP-1 drug pipeline.

They need forecasts at two horizons:

- Short-term (3 months) — quarterly position sizing
- Long-term (12 months) — annual allocation decisions

Response: Monthly log-adjusted closing price of LLY (Jan 2015 – Apr 2026, 136 observations)

136

Monthly Observations

5

Models Compared

2

Forecast Horizons Compared

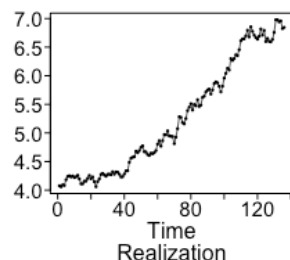
Variables: LLY (response) | XLV — Health Care Sector ETF | NVO — Novo Nordisk (GLP-1 competitor)

LLY LOG PRICE: NONSTATIONARY BEHAVIOR

REALIZATION

Clear Upward Trend

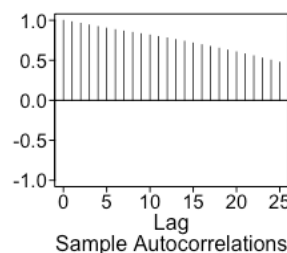
Persistent drift upward with no reversion to a fixed mean. Explosive acceleration from 2022 onward driven by GLP-1 drug approvals. Classic nonstationary wandering behavior.



ACF PLOT

Strictly Positive, Slow Decay

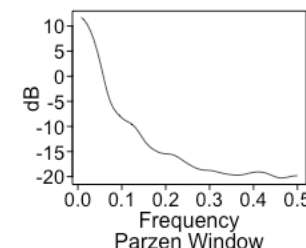
Autocorrelations are strictly positive and damp very slowly as lag increases — a damped exponential with no sinusoidal component. Hallmark of a $(1-B)$ factor.



SPECTRAL DENSITY

Sharp Spike at $f = 0$

All power concentrated at zero frequency, confirming the dominant upward trend. No evidence of cyclical or seasonal behavior in LLY's log price series.



After applying $(1-B)$, the series fluctuates around zero with no wandering — stationarity confirmed.

FOUR MODELS, TWO HORIZONS

01 ARMA(4,0)

Univariate · Differenced

- Fit to log returns (lly_diff)
- AIC: ARMA(0,0) best — returns $\approx$ white noise
- AR(4) used: nearly identical AIC (-5.144 vs -5.150)
- Model: $(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3 - \phi_4 B^4)X_t = a_t$

02 ARIMA(4,1,0)

Univariate · Log Price Level

- $d=1$ filter confirmed by EDA + overfit tables
- Forecasts on interpretable log price scale
- Model: $(1 - \phi_1 B - \dots - \phi_4 B^4)(1-B)X_t = a_t$

03 VAR(3) ×2

Multivariate · XLV and NVO

- VAR1: LLY + XLV (sector ETF)
- VAR2: LLY + NVO (direct GLP-1 competitor)
- VARselect $\rightarrow p=1$; used $p=3$ for flexibility

04 MLP Neural Net

Nonlinear · XLV Regressor

- $\text{difforder}=1$ to match ARIMA nonstationarity decision
- XLV lagged 1 month as external regressor
- 5 hidden nodes, 50 reps, median combination
- LLY lags 3 and 4 auto-selected internally

12-MONTH FORECASTS: LLY LOG PRICE

ARIMA(4,1,0) — Trend Projection

\$950 – \$1,254

RMSE: 0.492 (~49% error)

Works short-term. Projects explosive trend forward — intervals widen dramatically by month 12. Lower bound collapses to near zero.

- Short-term ASE: 0.004 ✓ Best 3-month
- Long-term ASE: 0.355 ✗ Struggles
- 95% PI at 12mo: \$3.94 – \$399,480 (reflects random walk uncertainty over 12 months)
- Honest reflection of random walk uncertainty

VAR1 (LLY + XLV) — Anchored Forecast

\$952 – \$1,140

RMSE: 0.212 (~21% error)

XLV anchors the forecast, preventing the explosive 2022–2024 trend from projecting forward indefinitely. More conservative and credible for client decisions.

- Short-term ASE: 0.070
- Long-term ASE: 0.045 ✓ Better than ARIMA
- 95% PI at 12mo: \$675 – \$1,926
- VAR2 (NVO) further improves to ASE: 0.037

MLP Best Overall: Long-term ASE 0.017 | Short-term ASE 0.025 | RMSE ≈ 0.13 (~13% error at 12 months)

MODEL PERFORMANCE COMPARISON

Model	3-Month ASE	12-Month ASE	3-Month RMSE	12-Month RMSE
ARMA (on log returns)	0.00602	0.010399	0.068	0.074
ARIMA(4,1,0) (on log price)	0.004055	0.355029	0.169	0.492
VAR1 (LLY + XLV)	0.069563	0.044851		
VAR2 (LLY + NVO)	0.061615	0.037499		
MLP (LLY + XLV, lag 1)	0.02489	0.016997		

ARMA is on the differenced scale (log returns) — not directly comparable to the others.
Comparison focuses on ARIMA, VAR1, VAR2, and MLP (all log price level).

FINAL RECOMMENDATION

3-Month Horizon

Quarterly Position Sizing

ARIMA(4,1,0)

Lowest static ASE (0.004). Captures random walk behavior effectively over short horizons. Simple and interpretable — easy to communicate to the client.

12-Month Horizon

Annual Allocation Decisions

MLP (XLV, lag 1)

Best ASE at 12 months (0.017). Adapts to nonlinear patterns and regime changes. VAR2 (LLY + NVO, ASE 0.037) is a strong interpretable alternative.

Key Takeaways

- LLY log returns behave close to white noise — limited predictable structure in monthly returns
- ARIMA best short-term: captures random walk behavior, lowest 3-month ASE (0.004)
- VAR2 (LLY + NVO) outperforms VAR1 — competitor adds more signal than the sector ETF
- Wide prediction intervals at 12 months reflect real equity uncertainty — communicate this to the client



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THANK YOU!

Questions & Feedback Welcome!